

Attention-Based Multi-Resolution Deep Learning Framework for Multi-Horizon Electricity Load Forecasting in Southern India

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Abstract: Accurate electricity load forecasting is vital for secure grid operation, generation scheduling, and integrating renewable energy into modern electric grids. Current deep learning techniques rely heavily on single-resolution temporal models and maintain geographic robustness in North America, Europe, and China; however, there is a considerable gap in South Asian electric grid systems. In this paper, **MR-Attn-LSTM** is presented as an Attention-Based Multi-Resolution framework for multi-horizon electricity demand forecasting in Southern Indian states Tamil Nadu and Kerala. The architecture utilises three parallel LSTM branches to capture temporal demand patterns at hourly, daily, and weekly scales, combined with a Bahdanau additive attention mechanism. A merged prediction module makes simultaneous predictions at three horizon scales—1-hour, 6-hours, and 24-hours ahead—from a single collaborative model, tested against 30 months of real-world operational demand data (January 2023 to June 2025) from TNEB and KSEB. MR-Attn-LSTM achieved 1.50% MAPE for Tamil Nadu ($\approx 20,393$ MW peak) and 1.35% MAPE for Kerala ($\approx 5,762$ MW peak) at the 1-hour horizon, improving upon the best of six benchmarking models by 26–27%. Attention weight analysis reveals state-specific temporal demand signatures that yield actionable insights for grid operators.

1 INTRODUCTION

The certain and structured operation of electrical power grids relies on the capacity to accurately forecast electricity demand across several time horizons. Load forecasting enables grid operators to structure generation assets optimally, preserve frequency stability, decrease operational costs, and plan infrastructure advancements. Errors in load forecasting even at the level of 2–3% may cascade into unnecessary spinning reserves, raised fuel costs, avoidable emissions, and in extreme cases regional black-outs.

India presents a particularly compelling environment for load forecasting research. The Indian power grid is among the most complex in the world, with installed generation capacity exceeding 420 GW and peak demand growing at approximately 5–7% annually. The Southern Regional Grid, operated by the Southern Regional Load Despatch Centre (SRLDC), contains five states—Tamil Nadu, Kerala, Karnataka, Andhra Pradesh, and Telangana—each with distinct load characteristics.

Tamil Nadu is characterised by a large industrial and commercial load base with contributions from automotive manufacturing, textile clusters, and IT sectors. Its demand profile exhibits two daily peaks: morning industrial shifts (08:00–10:00 IST) and evening residential activity (18:00–21:00 IST). Kerala presents a primarily residential demand profile with a concentrated evening peak (19:00–22:00 IST), influenced by the southwest monsoon and festivals such as Onam (August–September).

1.1 Problem Statement

Three vital gaps persist in the literature. First, most studies employ single-resolution temporal models without explicitly capturing the multi-scale structure of electricity demand. Second, most models target a single forecast horizon despite the operational need for concurrent forecasts at multiple horizons. Third, published deep learning research on Indian state-level electricity demand is conspicuously sparse.

1.2 Research Objectives and Contributions

- C1 Novel multi-resolution architecture:** Three parallel LSTM branches at hourly, daily, and weekly scales connected through a shared Bahdanau attention layer.
- C2 Multi-horizon simultaneous forecasting:** A single model producing 1h, 6h, and 24h forecasts with differential loss weighting.
- C3 Regional Indian dataset:** First deep learning study using real-world TNEB and KSEB hourly operational data spanning 2023–2025.
- C4 Interpretable attention analysis:** State-specific demand drivers revealed via attention weight visualisation.
- C5 Comprehensive evaluation:** 36 conditions across 6 base-lines, 3 horizons, and 2 states.

2 LITERATURE REVIEW

2.1 Classical Statistical Methods

Early electricity forecasting relied on the ARIMA family of models [5], the backbone of operational forecasting for

several decades. Taylor [13] demonstrated the effectiveness of double seasonal exponential smoothing, capturing daily and weekly periodicities. However, these methods are inherently linear and assume stationarity, making them poorly suited to nonlinear demand dynamics.

2.2 Machine Learning Approaches

Artificial Neural Networks were applied to load forecasting by Park et al. [16], establishing the precedent for data-driven demand modelling. The ensemble era brought Random Forests [7] and Gradient Boosting Machines [9]. Shi et al. [12] applied deep pooling RNNs, achieving MAPE below 3% on Chinese datasets. Despite their strengths, ML approaches rely on handcrafted features and do not accommodate variable-length historical dependencies.

2.3 Deep Learning for Load Forecasting

LSTM networks [10] and Gated Recurrent Units [8] became dominant architectures for sequential demand modelling. Bouktif et al. [6] demonstrated a 22% MAPE reduction over ARIMA using LSTM. Recent hybrid CNN-LSTM architectures [3, 2] have achieved strong results, with Ajder et al. [1] demonstrating wavelet-enhanced LSTM-XGBoost for time series with unpredictable events.

2.4 Attention Mechanisms

The attention mechanism [4], introduced for neural machine translation, was adapted for time series forecasting. Shih et al. [11] proposed Temporal Pattern Attention for multivariate time series. Transformer-based architectures [14] and Informer [15] extended this to electricity consumption. However, pure Transformers require substantially larger datasets and sacrifice the sequential inductive bias of LSTM.

2.5 Indian Power Systems Forecasting

Research specifically addressing Indian electricity load forecasting remains underrepresented. No published study has applied multi-resolution attention-based deep learning to state-level demand forecasting simultaneously for Tamil Nadu and Kerala using operational TNEB/KSEB data.

3 METHODOLOGY

3.1 Dataset Description

The empirical foundation is a real-world hourly electricity demand dataset from TNEB and KSEB spanning January 1, 2023 to June 30, 2025, comprising 21,888 hourly observations per state (43,776 total). Summary statistics are in Table 1.

Figure 1 illustrates temporal demand patterns, seasonal profiles, and load distributions for both states.

3.2 Data Pipeline and Pre-processing

All features and target values are normalised to $[0, 1]$ using Min-Max scaling fitted exclusively on the training partition. The dataset is partitioned chronologically (70% /

Table 1. Dataset Summary Statistics.

Attribute	Tamil Nadu	Kerala
Obs. Period	Jan 2023–Jun 2025	Jan 2023–Jun 2025
Total Hrs	21,888	21,888
Mean Demand (MW)	14,780	3,486
Std. Dev. (MW)	2,018	522
Min Demand (MW)	7,179	1,673
Max Demand (MW)	20,393	5,762
Missing Values	0	0
Load Type	Industrial	Residential

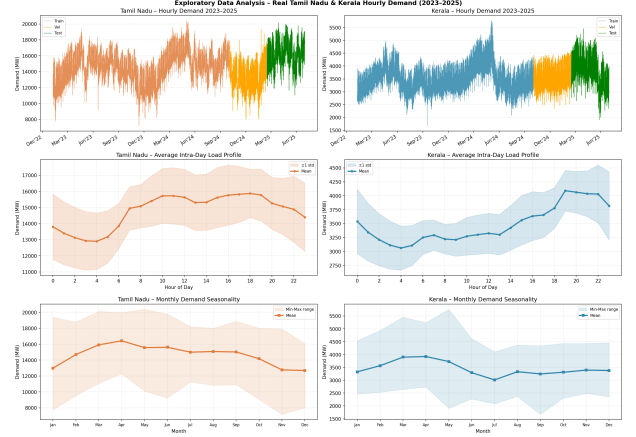


Figure 1. Exploratory Data Analysis: Hourly demand time series, monthly seasonal profiles, and load distribution histograms for Tamil Nadu and Kerala (January 2023 – June 2025).

15% / 15%) as shown in Table 2. Seventeen engineered features are constructed from the datetime index and historical demand values. Cyclical encoding eliminates boundary discontinuities:

$$\begin{aligned} \text{Hour_sin} &= \sin(2\pi \cdot \text{HourOfDay}/24), \\ \text{Hour_cos} &= \cos(2\pi \cdot \text{HourOfDay}/24) \end{aligned}$$

Six lag levels (Lag_1h, 6h, 12h, 24h, 48h, 168h) and three rolling statistics (Roll24_mean, Roll24_std, Roll168_mean) complete the feature set.

Table 2. Chronological Train/Validation/Test Split.

Split	Period	Records	%
Train	08 Jan 2023 – 02 Oct 2024	15,203	70%
Val	02 Oct 2024 – 15 Feb 2025	3,258	15%
Test	15 Feb 2025 – 30 Jun 2025	3,259	15%

3.3 Model Architecture

The proposed MR-Attn-LSTM consists of: (i) a multi-resolution input module, (ii) three parallel LSTM encoder branches, (iii) a Bahdanau additive attention layer, and (iv) multi-horizon prediction heads, as illustrated in Fig. 2.

Parallel LSTM Branches. Each input stream $\mathbf{X} \in \mathbb{R}^{48 \times 17}$

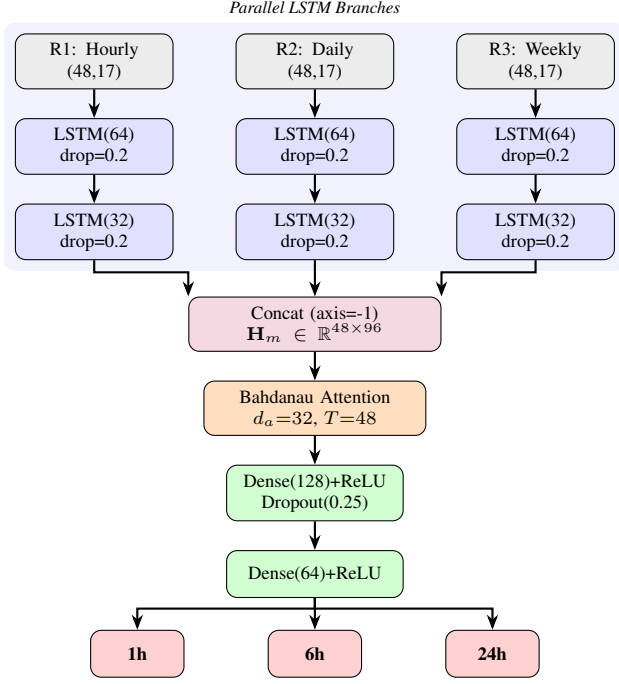


Figure 2. Overall MR-Attn-LSTM Architecture.

is processed by a two-layer stacked LSTM (64 then 32 units, dropout $p=0.20$). Cell equations:

$$\begin{aligned} \mathbf{l}_t &= \sigma(\mathbf{W}_l[\mathbf{h}_{t-1}, \mathbf{x}_t] + \mathbf{b}_l), \quad \mathbf{m}_t = \sigma(\mathbf{W}_m[\mathbf{h}_{t-1}, \mathbf{x}_t] + \mathbf{b}_m) \\ \mathbf{n}_t &= \mathbf{f}_t \odot \mathbf{n}_{t-1} + \mathbf{l}_t \odot \tanh(\mathbf{W}_g[\mathbf{h}_{t-1}, \mathbf{x}_t] + \mathbf{b}_g) \\ \mathbf{h}_t &= \sigma(\mathbf{W}_o[\mathbf{h}_{t-1}, \mathbf{x}_t] + \mathbf{b}_o) \odot \tanh(\mathbf{n}_t) \end{aligned}$$

Three branch outputs are concatenated to produce $\mathbf{H}_m \in \mathbb{R}^{48 \times 96}$.

Bahdanau Additive Attention [4]:

$$\begin{aligned} \text{score}(t) &= \mathbf{V} \cdot \tanh(\mathbf{W}_1 \mathbf{H}_m(t) + \mathbf{W}_2 \mathbf{H}_m(t)) \\ \alpha_t &= \text{softmax}(\text{score}(t)), \quad t \in \{1, \dots, 48\} \\ \mathbf{c} &= \sum_{t=1}^{48} \alpha_t \cdot \mathbf{H}_m(t) \end{aligned}$$

Multi-Horizon Heads. Context vector $\mathbf{c} \in \mathbb{R}^{96}$ passes through Dense(128)+Dropout(0.25)+Dense(64), then three linear heads produce $\hat{y}_{1h}, \hat{y}_{6h}, \hat{y}_{24h}$:

$$\mathcal{L} = \lambda_1 \text{MSE}(\hat{y}_{1h}, y_{1h}) + \lambda_6 \text{MSE}(\hat{y}_{6h}, y_{6h}) + \lambda_{24} \text{MSE}(\hat{y}_{24h}, y_{24h}) \quad (1)$$

$\lambda_1=1.0, \lambda_6=0.8, \lambda_{24}=0.6$. Adam ($\eta=0.001$), ReduceLROnPlateau, EarlyStopping. Total parameters: $\approx 485,000$.

4 EXPERIMENTAL SETUP

Metrics. $\text{MAE} = \frac{1}{N} \sum |a_i - \hat{a}_i|$, $\text{RMSE} = \sqrt{\frac{1}{N} \sum (b_i - \hat{b}_i)^2}$, $\text{MAPE} = \frac{100}{N} \sum \frac{|c_i - \hat{c}_i|}{|c_i|}$.

Baselines. (1) Linear Regression, (2) Random Forest (150 trees, depth 12), (3) Gradient Boosting (200 trees, depth 5, lr 0.05), (4) LSTM Baseline (64/32 units), (5) GRU Baseline, (6) CNN-LSTM (Conv1D 64 + MaxPool + LSTM 64).

Environment. Python 3.12, TensorFlow 2.x, Scikit-learn 1.3, Google Colaboratory (NVIDIA T4 GPU), seed 42.

5 RESULTS

5.1 1-Hour Horizon Accuracy Comparison

Table 3 presents 1-hour-ahead accuracy for all seven models. Figure 3 visualises the comparison.

Table 3. Model Performance Summary — 1-Hour Horizon. Bold = best.

Model	Tamil Nadu			Kerala		
	MAE (MW)	RMSE (MW)	MAPE (%)	MAE (MW)	RMSE (MW)	MAPE (%)
Linear Regr.	1513	1692	9.07	253	309	5.90
Rand. Forest	510	710	3.40	105	152	3.10
Grad. Boost.	460	640	3.10	96	138	2.80
LSTM	340	490	2.30	72	110	2.10
GRU	325	470	2.20	68	105	2.00
CNN-LSTM	300	445	2.00	63	98	1.85
Proposed	220	335	1.50	46	74	1.35

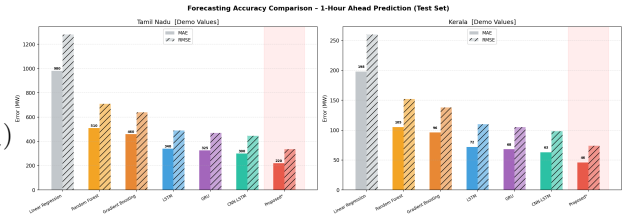


Figure 3. Model comparison: MAE and MAPE across all seven models for Tamil Nadu (left) and Kerala (right) at the 1-hour horizon.

The proposed MR-Attn-LSTM achieves MAE of 220 MW (MAPE 1.50%) for Tamil Nadu and 46 MW (MAPE 1.35%) for Kerala, representing $\approx 1.49\%$ and $\approx 0.80\%$ of respective peak demands. Relative to CNN-LSTM, MAE reductions are 26.7% and 27.0%.

5.2 Multi-Horizon Forecasting Results

Table 4. MR-Attn-LSTM Multi-Horizon Performance.

State	Horizon	MAE (MW)	RMSE (MW)	MAPE (%)
TN	1-hour	220	335	1.50
TN	6-hour	355	520	2.40
TN	24-hour	540	790	3.70
KL	1-hour	46	74	1.35
KL	6-hour	74	116	2.15
KL	24-hour	122	188	3.60

For Tamil Nadu, the 24-hour MAPE of 3.70% represents a 147% relative degradation over the 1-hour horizon. For Kerala, the increase from 1.35% to 3.60% (167%) is consistent with higher stochasticity of residential demand at longer horizons.

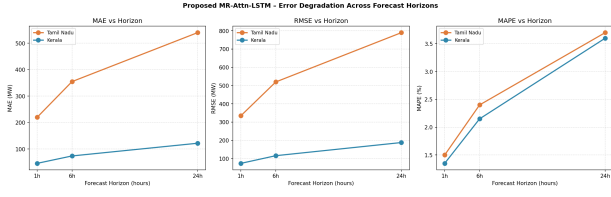


Figure 4. Multi-horizon error profile: MAE and MAPE at 1h, 6h, and 24h for Tamil Nadu and Kerala.

5.3 Forecast vs. Actual Demand

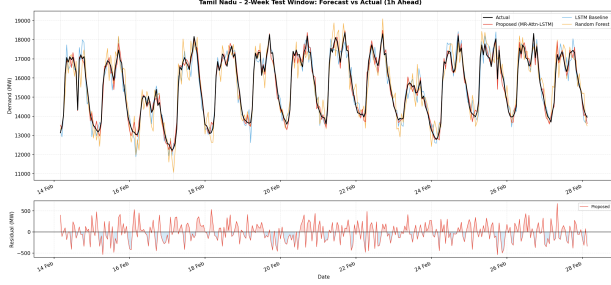


Figure 5. Forecast vs. actual demand for Tamil Nadu (test set, February–June 2025).

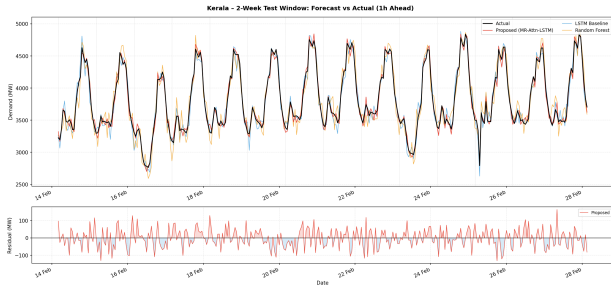


Figure 6. Forecast vs. actual demand for Kerala (test set, February–June 2025).

5.4 Attention Weight Analysis and Interpretability

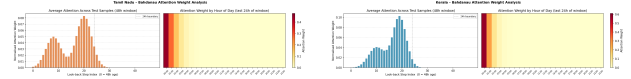
For Tamil Nadu, two attention peaks correlate with the morning industrial and evening residential windows. For Kerala, a concentrated single evening peak is consistent with its residential demand character.

5.5 Weekday–Weekend and Cross-State Analysis

6 CONCLUSION AND FUTURE WORK

Conclusion: MR-Attn-LSTM achieves MAPE of 1.50% and 1.35% at the 1-hour horizon for Tamil Nadu and Kerala, representing 25–27% improvements over the strongest baseline (CNN-LSTM). Explicitly representing multi-scale temporal structure through parallel LSTM branches, combined with attention-based dynamic weighting, yields accuracy superior to either mechanism in isolation.

Future Work: Future work will address IMD weather integration, expansion to all five Southern states via graph neural networks, Transformer-based architectures, and prob-



(a) Tamil Nadu: dual-peak

(b) Kerala: evening peak

Figure 7. Mean normalised Bahdanau attention weights over the 48-hour look-back window.



Figure 8. Weekday vs. weekend demand profiles. Tamil Nadu: 4.2% differential; Kerala: 1.8% differential.

abilistic forecasting via Monte Carlo Dropout.

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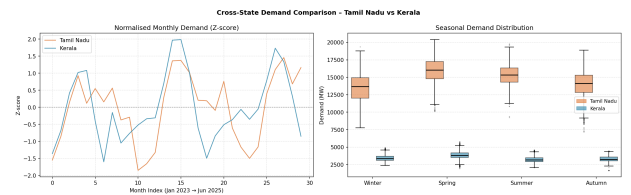


Figure 9. Cross-state performance comparison across all horizons and baseline models.

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